PRACTICAL – 5

Registration Form with Frontend Validation and User-Friendly Error Handling

Aim: To create a student registration form using HTML5 and JavaScript with frontend validation for name, email, mobile number, password, confirm password, course, year, gender, and terms acceptance using Regular Expressions and user-friendly error messages.

echnologies Used

- **HTML5**
- **CSS3**
- **JavaScript ES6**
- **Regular Expressions**
- **Browser Developer Tools**

Implementation

1. DOM Selection

JavaScript selects form elements using DOM methods.

Example:

let name = document.getElementById("name").value;

This retrieves the value entered by the user.

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2. Regular Expression Validation

Regular Expressions are used to check whether the entered data follows the required format.

Name

/^[A-Za-z]{2,}$/

It allows letters and spaces.

Mobile Number

/^[0-9]{10}$/

It accepts exactly 10 digits.

Email

/^[^\s@]+@[^\s@]+\.[^\s@]+$/

It checks the basic structure of an email address.

Password

/^(?=.*[A-Za-z])(?=.*\d).{8,}$/

It requires at least 8 characters, including a letter and a number.

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3. Password Confirmation

The password and confirm password fields are compared.

```
if (password !== confirmPassword) {
   document.getElementById("confirmPasswordError").textContent =
      "Passwords do not match.";
}
```

An error message is displayed when the two passwords are different.

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4. Password Strength Meter

The password strength is checked while the user types.

```
if (password.length < 6) {
   strength.textContent = "Weak Password";
} else if (password.length < 8) {
   strength.textContent = "Medium Password";
```

} else {

strength.textContent = "Strong Password";

}

This provides immediate feedback to the user.

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5. Form Submission

The submit event is used to validate the form before submission.

form.addEventListener("submit", function(event) {

event.preventDefault();

// validation

});

preventDefault() prevents the form from submitting until the validation is completed.

